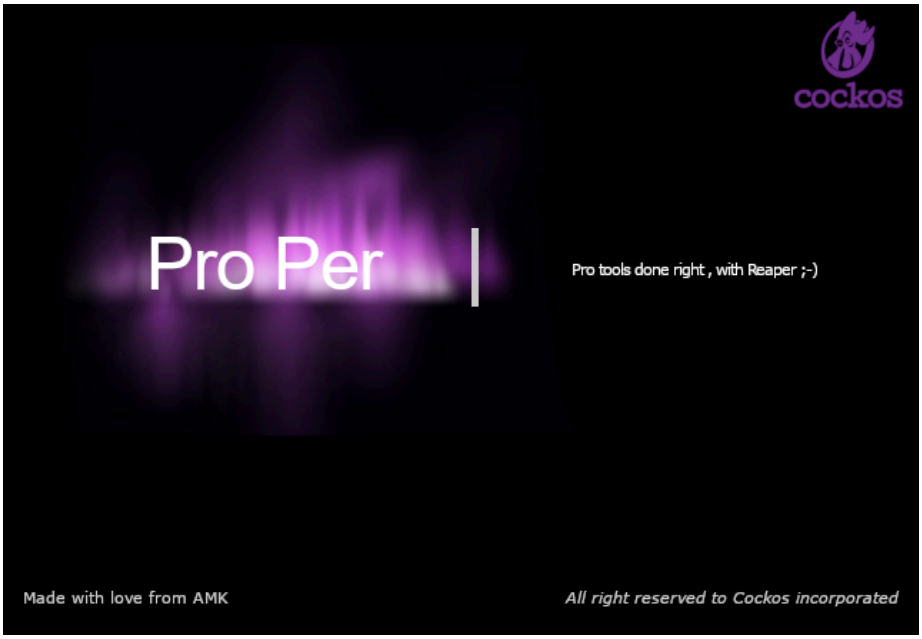
# PRO PER (Beta 1.7.83) – USER MANUAL




## 1. SYSTEM ARCHITECTURE & TRACK ROUTING

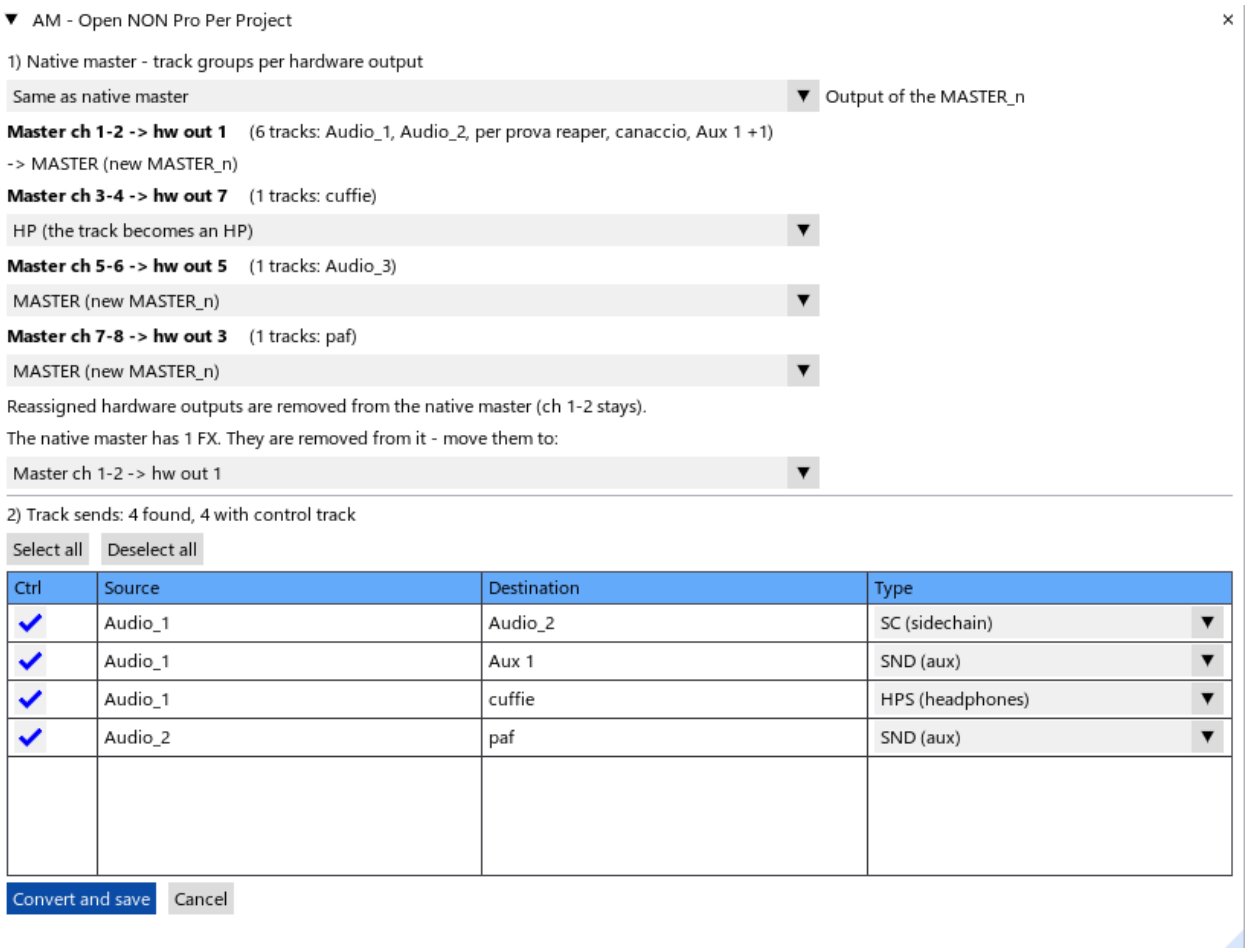
### System Requirements & Configuration Limits

- **Language Constraint:** REAPER's interface language must be strictly set to English. The integrated custom scripts rely on English string parsing; using other language localization packages will prevent macro execution and break script states.
- **UI/UX Scaling Framework:** The minimum resolution is 1920 x 1080p.The default Pro Per theme is self-adapting (stretched). For HiDPI resolutions, the configuration automatically installs the Pro Per_150 and Pro Per_200 themes separately to ensure optimal loading speed when launching Reaper.
- **Mandatory Extensions:** Proper processing and modal window operations require the prior installation of the following packages via ReaPack: SWS Extension, JS_ReaScriptAPI, and ReaImGui.

### Open Non Pro Per Project script

- **Native Session Integration:** Allows opening projects created with standard (native) REAPER versions. It automatically adapts the session to ensure full compatibility across all other features and structures managed by Pro Per, preventing routing conflicts and system inconsistencies associated with legacy track setups.

- **Selective Master/Control Track Assignment:** To maintain compatibility with Pro Per's internal architecture, the system processes existing sends within the native project. Upon opening, the user must explicitly choose which sends will generate dedicated hidden Control Tracks and floating fader anchors, ensuring that the session routing framework remains organized and optimized.Similarly, if there are multiple hardware outputs in a project—whether on Reaper's native Master or on a track—you will need to choose whether to assign it to a new Master or to an HP Master.



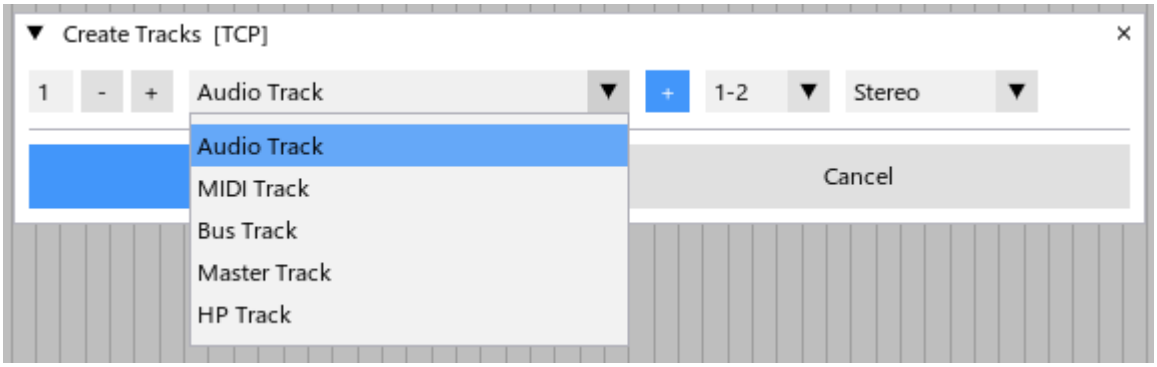

## Master Track Audition Path Behavior

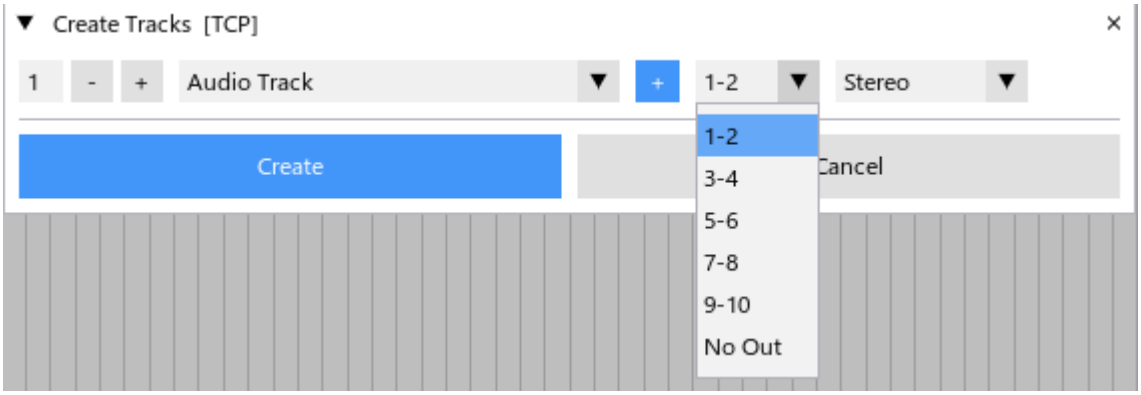
- **Master Track Routing Specialization:** In Pro Per, ***REAPER's native Master Track*** *is utilized strictly as an isolated audition path rather than a standard session summing bus*. It can be assigned to any specific hardware output and handles exclusively the monitoring of track scrub operations and the solo audio streams going to the artists' headphone cue masters. Additionally, this track serves as the *visual housing for the project Tempo Map*.
- **In-The-Box (ITB) Mixing Layout:** When mixing entirely in the box, the user can limit the session layout to a single Master fader. This fader is automatically initialized and deployed upon creating the first track from the track creation window.

## Track Creation Terminal

- **Track Creation Window:** Pressing Cmd + Shift + N (macOS) or Ctrl + Shift + N (Windows) opens the custom New Track window. This terminal allows for the immediate generation and configuration of Audio, MIDI, Bus, Master, HP (Headphone Master) Tracks, and Surround configurations.
  When the Window is open:
  change between types of track: ctrl+left arrow, ctrl+right arrow (macOS) or  Win+left arrow, Win+right arrow (Windows).
  add a new line: ctr+N (macOS) , Win+N (Windows).

- **MIDI Track Profile:** Automatically initialized with a layout and interface tailored for Instrument tracks and virtual instrument handling.
- **Bus Track Profile:** Replicates the behavior of a Pro Tools Aux Input, optimized for FX returns and secondary group processing. For master sub-group summing, routing folders with bespoke layouts are recommended instead.
- **HP Master Cue Bus Track Profile:** Generates dedicated HP masters that could be mapped to specific physical interface outputs. It receives the dedicated send assigned and features a dedicated Solo button labeled “Hs”, scripted to monitor the artist cue mixes without interruption of the audio streams to other HPs Cue Masters.
- **Multi-Channel Surround Formats:** Includes built-in configuration settings for 5.1 Surround and 7.1 Surround track profiling. It structures output streams into discrete hardware channel groups (blocks of 6 for 5.1, blocks of 8 for 7.1) and deploys multi-send tracking matrices directly to a single surround master.




Create Tracks [TCP]
1
-
+
Audio Track
+
1-2
Stereo
Stereo
5.1 Surround
7.1 Surround
7.1.2 Surround
7.1.4 Surround
Create


AUTO
auto off
Inserts
center
S
M
HS
0.00
pha
Audio_1
MIDI_1
Aux 1
HP 1
MASTER_1
track 1
track 2
track 3
track 4
track 5
master
snd rcv

## Video Import & Window Management

- **Native Video Integration:**

  Importing a video file from any source automatically processes and separates the video stream and its accompanying audio onto two distinct, dedicated tracks, mimicking the standard Pro Tools workflow. To ensure seamless synchronization during arrangement and editing, the resulting video and audio timeline items are automatically grouped together.

- **Dynamic Video Info Display:**

  The custom video track header (TCP) dynamically displays essential metadata directly on the interface, including the video's resolution width and height (e.g., 1920x1080) and its active frame rate (FPS), ensuring instant visual verification without opening secondary property windows.

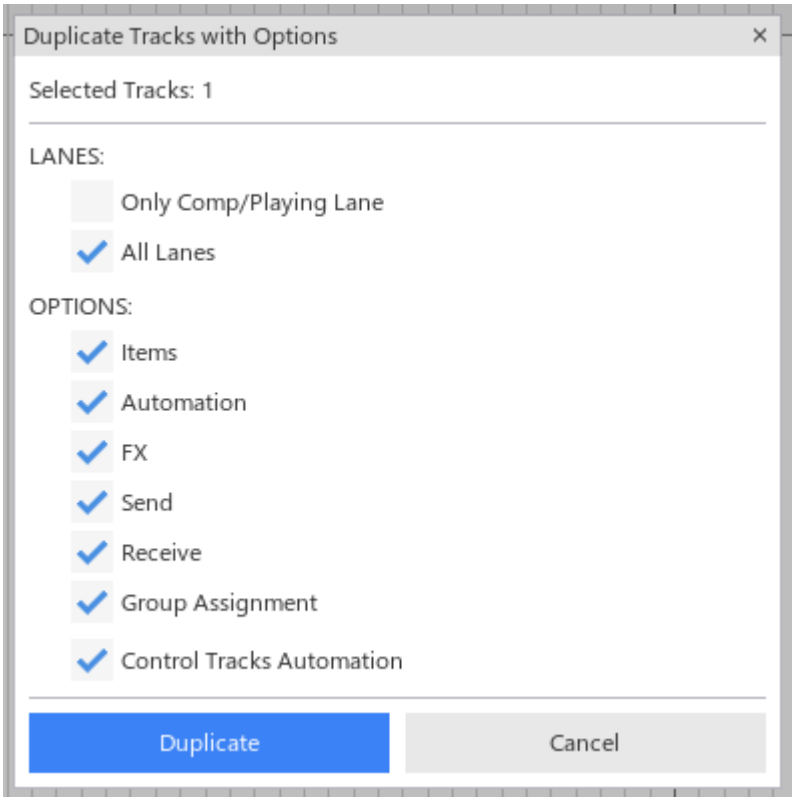
- **Dedicated Video Window Toggle:** Pressing Cmd + 9 (macOS) or Ctrl + 9 (Windows) on the numeric keypad instantly toggles the visibility of the dedicated floating Video Window, allowing for quick preview adjustments and workspace layout optimization.




## Duplicate Tracks with Options

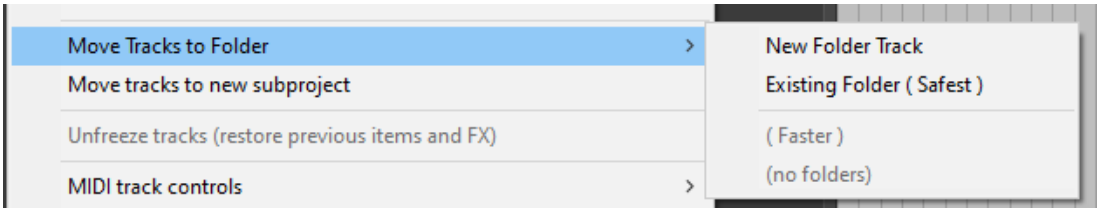
- **Duplicate Tracks Window:** Selecting one or more tracks and triggering the duplicate command opens the custom **"Duplicate Tracks with Options"** popup window. Every duplicated track automatically has the suffix `_dup` appended to its name.
- **Lanes Options:**
  - **Only Comp/Playing Lane:** Duplicates exclusively the active composite or playing lane of the source track, ignoring hidden take layers.
  - **All Lanes:** Duplicates the entire hierarchical structure of the Fixed Item Lanes, preserving all underlying take history and reference tracks.

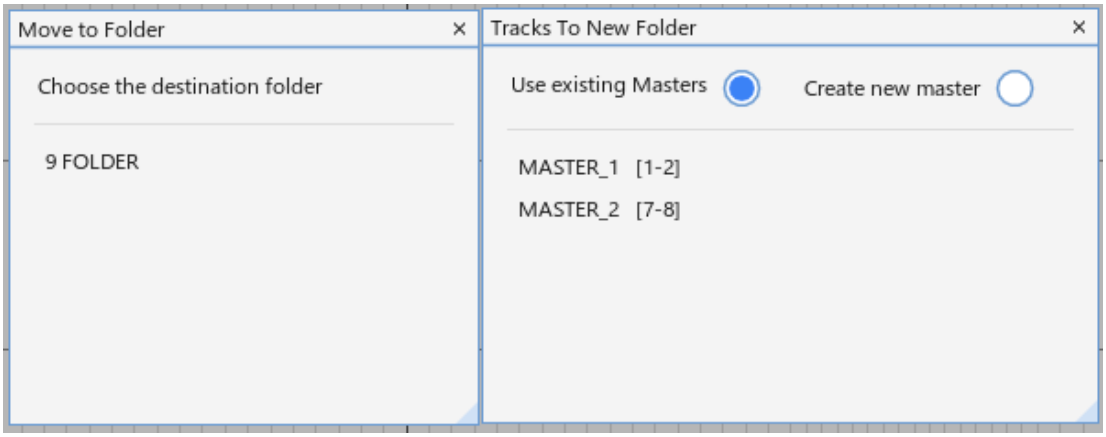
- **General Copy Options:**
    - **Items:** Copies all audio and MIDI items present on the timeline of the source tracks.
    - **Automation:** Copies standard track and item automation envelopes.
    - **FX:** Copies all plugin insert chains and processing units.
    - **Send:** Copies the internal routing sends.
    - **Receive:** Copies the internal routing receives.
    - **Group Assignment:** Replicates track group linking and parameter association parameters.
    - **Control Tracks Automation:** (Track Control Automation) Decides whether or not to copy the automation data residing specifically on the hidden Control Tracks.

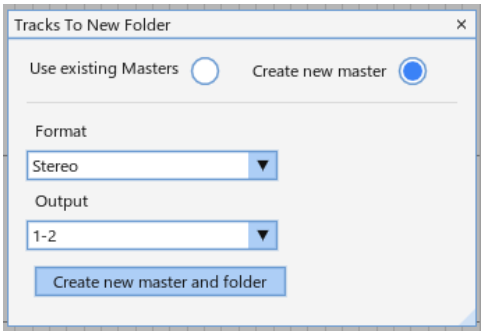



## Advanced Folder Generation and Hardware Routing

- **Contextual Routing Script:** The native REAPER folder creation layout has been replaced with a contextual routing script. Selecting a multi-track ran7ge and opening the right-click menu allows the user to either move the selection to an existing folder structure, or generate a new folder track instantly like natively in Reaper but managing the internal Master bus outputs structure of Pro Per
- **Direct Routing and I/O Assignment:** The generation window features options to route the newly created folder directly to an existing session Master bus or create a new Master and route it to hardware outputs directly inside the same prompt.




## Control Track Send Management System

- **Automated Send Architecture:** Creating an internal send (via drag-and-drop or Shift-clicking the send drop-down menu, or via the tracks menu through a dedicated entry “Send Track to Aux/HP/Master”), generates a dedicated hidden Control Track in the background:
  **SND** for ***Send*** (Post fader default), **SC**, for ***Sidechain*** (Pre fader default), **HPS** for ***Headphones*** (Pre fader default).
  The created Control Tracks have a customization of native TCP and MCP layout with buttons for Pre/Post Fader and Stereo/Mono switching.
  The resulting floating fader popup acts as an anchor to control this hidden layer, keeping REAPER's standard layout unencumbered by excessive floating windows.
  For sidechain routing, the supported mode is by dragging directly on the plugin.
- **Fader Popup Functionality:** Built using Lua and ReaImGui (with Atlas rendering), this independent send fader includes buttons for Mute, Solo (auditioning exactly what is feeding the destination bus), and Pre/Post Fader switching. It also includes a Stereo/Mono button addressing source track sends to manage panning behaviors on sends.
  - **Automation:**

Send automation modes can be cycled directly (Trim/Off, Read, Touch, Latch, Write) from the floating popup fader window. Envelopes are written directly to the control layer and can be displayed using Ctrl + W, which shows the hidden Control Track.

- **Arrange Window Fast Opening System:**
  You can open a ***Send/Sidechain*** Floating Fader just by double clicking on the Tcp Track Header of the Source Tracks.
  All Send Faders for that track will be shown.
  Similarly you can open the ***HPS*** send Floating Fader by Alt + double click on the TCP Track Header of the Source Tracks.
- **Naming Sync:**
  Naming is bi-directionally dynamic: modifying the name of a source track or destination bus automatically updates the control track identifiers.
- **Automatic Synchronization:** The Control Track and the send on the source track are inextricably linked. By deleting a Control Track, the script automatically removes the corresponding send, preventing the accumulation of orphan or residual routing in the project. Similarly, removing the send automatically cleans up the control track.
- **Automatic Send and Control Track Purge:** When deleting an Aux (Bus) track, the system automatically cleans up and deletes any associated Sends and Control Tracks assigned to it, unless those same sends are also routed to other active Aux tracks.
- **Dynamic Color-Coding:** Control tracks feature fixed, theme-specific coloration rules based on their operational profile, both on Track header and in Fader.
- **Control Track Magnetism:** Control tracks are bound to their source tracks in the edit and mix window; moving an audio track automatically updates its position so its corresponding send layers move along directly behind it.
  This is particularly important on the mixer window.
  Although you can open in the mixer view a floating fader from the left click menu, it is faster to use the **Ctrl+W** or **Ctrl+E** shortcut to show Control Tracks right sided to their Source Tracks and operate directly from there.
- **User-Defined Parallel Mixers:** This custom engine allows you to select any group of tracks and deploy multiple parallel mixer windows simultaneously. This lets you isolate and mix sub-groups of tracks in detached visual containers without changing your main arrangement views.

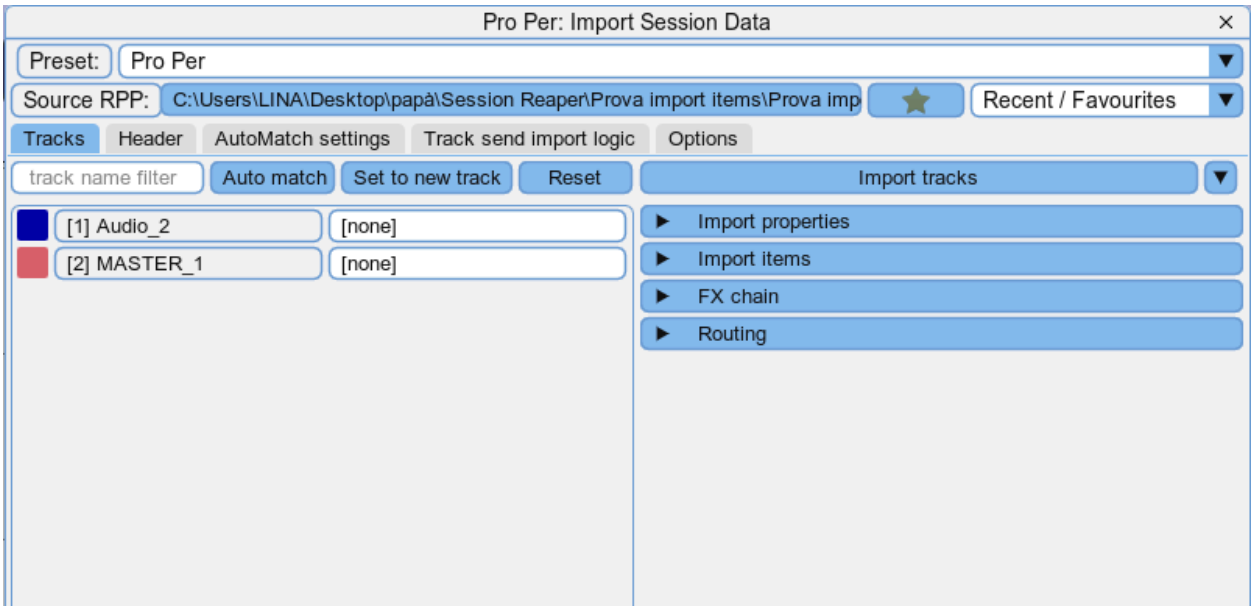



## Import Session Data (MPL Fork Customization):

- **Customized Session Data Import:**
  To guarantee full compatibility with Pro Per's internal architecture, the native session import routines have been replaced with a dedicated fork of MPL's Import Session Data script.

- **Framework Preservation:**
  This customized tool allows you to seamlessly import tracks, media items, routing matrices, and plug-in chains from other projects while automatically respecting Pro Per's proprietary

structures—such as hidden *Control Tracks*, specialized *Bus Track Profiles*, and headphone cue mappings—preventing any structural routing corruption during the import process.




# 2. WORKSPACE & WINDOW MANAGEMENT

## Transport Layout & Theme Adjuster

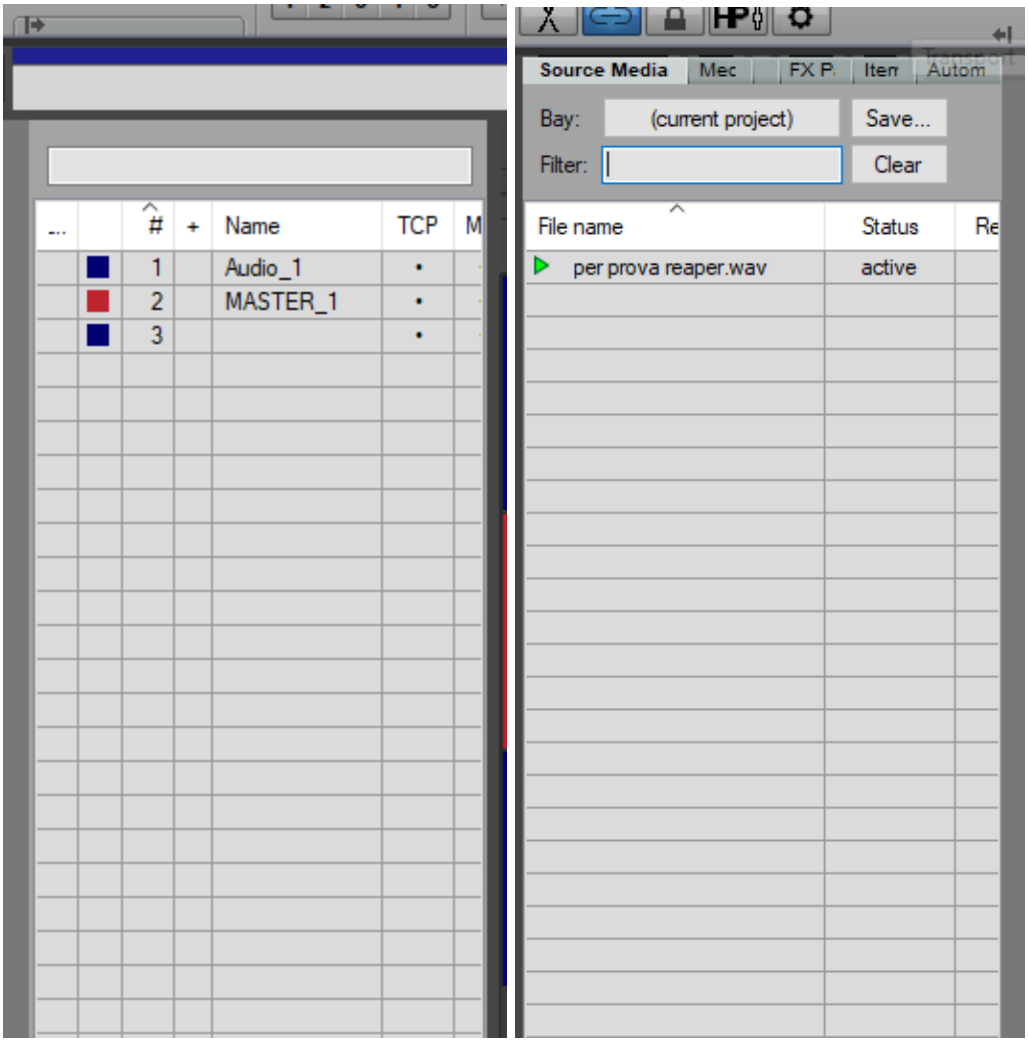
- **Transport Bar Customization:** The main transport bar has been entirely customized to replicate the layout and user experience of Pro Tools as closely as possible. To maximize workspace efficiency, additional functionalities have been directly integrated into the transport to **minimize the need for secondary toolbars**. By clicking the dedicated button to open the **Theme Adjuster** and modifying the **"Transport Hide Sections"** slider, users can dynamically show or hide specific modules, tailoring this centralized hub to their exact tracking or mixing requirements.



## Side Panels Toggle Buttons

- **Track and Group Manager Docking:** Located on the top-left section of the main transport bar. Clicking the left arrow button instantly opens or closes the custom Track Manager and Group Manager panels on the left side of the screen, matching the Pro Tools layout.

- **Clip List Integration:** Located on the top-right section of the main transport bar. Clicking the right arrow button toggles the modified Project Bay panel on the right side of the screen. This panel functions similarly to the Pro Tools Clip List, showing all audio files, takes, and media items currently loaded into the session.




## Zoom Controls

- **Horizontal Zoom Presets:**

  Five dedicated numeric buttons (1 to 5) are placed on the transport area to recall specific horizontal magnification levels instantly. To store a custom zoom level into one of the slots, hold the Cmd key (macOS) or Ctrl key (Windows), and click on the preferred number on the keyboard.

- **Zoom Toggle:**

  Pressing the single key E centers and magnifies the boundaries of the currently selected audio item, filling the timeline view for detail work. Pressing E again restores the previous timeline zoom level.

- **Zoom Waveform, Zoom MIDI Tracks, and Zoom Horizontally In & Out Transport Buttons:**

These buttons control the zoom as indicated by their names, using the same shortcuts used in Pro Tools.




## Numeric Keypad Mapping

- **Dedicated Numeric Keypad Transport Mode:**
  The numeric keypad features a full mirror mapping of traditional Pro Tools transport functions for every single key.
  • **Key 0**: Play / Stop toggle.

  • **Key 1:** Rewind (moves backward by one full measure/grid line; holds for continuous rewind).

  • **Key 2:** Fast Forward (moves forward by one full measure/grid line; holds for continuous fast forward).

  • **Key 3:** Record Enable transport toggle.

  • **Key 4:** Loop Playback mode toggle.

  • **Key 5:** Autopunch time selection recording mode toggle.

  • **Key 6:** QuickPunch recording mode toggle.

  • **Key 7**: Click / Metronome on/off toggle.

  • **Key 8**: Count off toggle.

  • **Key 9**: MIDI Merge / Replace mode toggle.

  • **Key [*] (Asterisk):** Triggers direct location/timecode entry.

  • **Key [.] (Period):** Jump to Marker. Pressing the period key followed by a specific marker number instantly moves the edit cursor directly to that marker on the timeline.

  [.]enter number[.]

- **Latch Mode Toggle:**
  A dedicated button on the main transport bar allows the user to enable or disable Latch Mode for Fast Forward (Key 2) and Rewind (Key 1) functions. This determines whether continuous scanning latches upon a single press or requires the key to be physically held down.

  When Latch Mode is disabled (Off), a single key press of *Key 1 (Rewind)* or *Key 2 (Fast Forward)* will move the edit cursor backward or forward by one measure (or less, depending on the current grid unit and zoom/view settings).



## Emulated True QuickPunch Mode

To achieve a genuine Pro Tools-style QuickPunch behavior, Pro Per utilizes a highly customized framework built exclusively around REAPER's ***Fixed Item Lanes engine***. This architecture guarantees non-destructive background capture during both punch-in and punch-out operations while running continuous playback: ***the audio is permanently recorded in background*** from the exact moment Play is engaged until Stop is pressed.

- **Real-Time Input Monitoring Behavior (Tape Auto Style)**
  The monitoring mirrors the response of the Pro Tools engine, automatically tracking the transport state for any record-armed channel:

  ○ **During Playback (Pre-Punch):**
  You will hear exclusively the audio items already residing on the timeline. The live physical input (microphone/instrument) is completely muted, allowing the artist to listen to their previous performance.

  ○ **At Punch-In:** Triggered instantly by pressing **Key 3** on the numeric keypad (or Cmd+Space / Ctrl+Space). Playback of the existing timeline audio on that specific channel is immediately cut off, and the monitoring switches to **100% live input** in real time.

  ○ **At Punch-Out:** Pressing Key 3 again instantly restores timeline playback, seamlessly unmuting the recorded track material and silencing the live input monitor.

- **Input Monitor Modes**
  Depending on the session requirements, the engineer can cycle the track monitoring configuration to adapt to different tracking workflows:

  ○ **Tape Mode:** The traditional behavior described above (live input is audible only during active punch-in segments).

  ○ **Input Only Mode:**The live physical input remains permanently audible in the artist's cue mix during playback, pre-roll, and recording, regardless of the punch states.

- **Direct Item Trimming (Main Workflow):**
  Because the full background pass is embedded directly into the recorded clips, you do not need to expand the lanes to recover item handles. Even when lanes are collapsed, you can instantly reveal the audio recorded before the punch-in and after the punch-out by simply performing standard edge trimming (Trim Forward/Backward) directly on the main punch items in the timeline.
- **Activation:**
  Toggle the QuickPunch mode on or off via the dedicated transport button or by pressing ***Key 6*** on the numeric keypad. This automatically forces the recording engine into *Auto-Punch Selected Items* behavior.
- **Multi-Track & Grouped Execution:**

The QuickPunch engine natively supports ***multi-track tracking and grouped track*** behaviors. Performing a punch operation on a grouped track instantly executes the punch-in/out and background capture across all linked tracks simultaneously, preserving strict phase alignment.

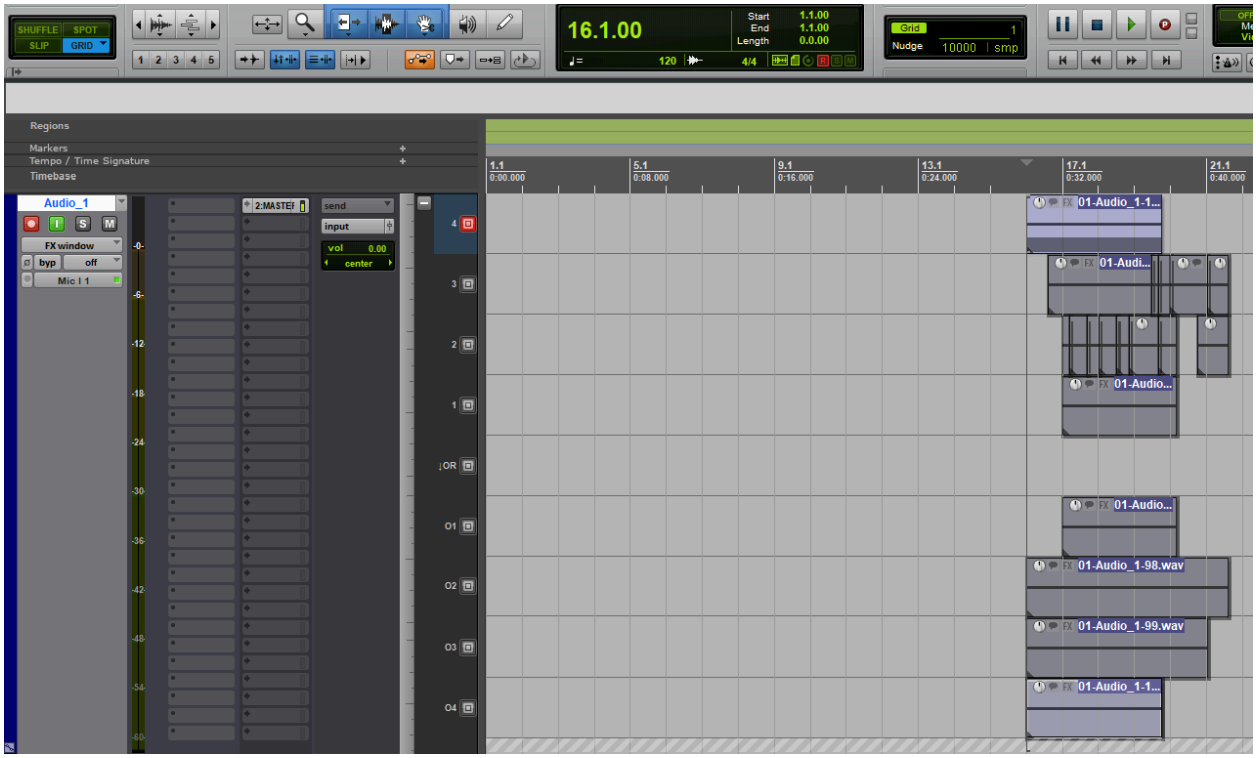
- **Playlist Workflow Integration:**
  Pro Tools traditional shortcuts for *New Playlist* and *Duplicate Playlist* are fully mapped and integrated here to manage the generation of ***New and Duplicate Lanes***.
  New Lanes: Ctrl + \ (Mac) or Start + \ (Windows)
  Duplicate Lanes: Command + Ctrl + \ (Mac) or Ctrl + Start + \ (Windows)
- **Visual Architecture & Safety Layers:**
  Expanding the Fixed Item Lanes reveals the full history of the recording session, structured for standard non-destructive editing and advanced asset recovery:
  - ***Punch Takes (Upper Lanes):***

    Displays all individual punch-in/punch-out segments as are recorded. Just like in Pro Tools, these items are completely non-destructive.

  - ***The "OR" Reference Lanes (Lower Lanes):***

    Separated below by an empty placeholder lane named !OR, dedicated lanes named sequentially ("O1", "O2", etc.) permanently house the complete, uncut, continuous audio files from each play-to-stop pass. These lanes act strictly as a secondary fallback and visual map, allowing the engineer to instantly reference or drag continuous central sections that would otherwise be cumbersome to recover via standard item edge-trimming. They are numerically inverted compared to the lanes above.

- **Numbering Conventions:**All Lanes are automatically numbered and structured following the native Pro Tools asset identification schema.
- **Punch-in / Punch-out**
  To Punch-in and Punch-out use **key 3** on numeric Keypad or Command + space ( MacOs ) / Ctrl + space ( Windows ).
- **Shortcuts for new lanes ( playlist ) and duplicate lanes:**
  Simplified shortcuts to: ctrl+\ , command +\ to duplicate; Option + \ ( both Windows and Mac Os ), for new lanes.




**CAUTION:** QuickPunch cannot be used to record tracks to tracks with Record Outputs rec modes

## Advanced Floating Parallel Mixers

- **Detached Sub-Group Mixing Architecture:**
  Born for Send Floating Faders, Pro Per features a custom Floating Mixer engine. Unlike Pro Tools, which restricts you to a single main mixer window or rigid window configurations, this tool allows you to select any arbitrary group of tracks and instantly deploy multiple, independent parallel mixer windows simultaneously.
  It also allows you to select automation mode and write automation for volume and pan or open/close folders directly from the Floating mixer.

  ***Workflow Advantages:***

  - Focused Isolation: Isolate specific sub-groups (e.g., Drums, Backing Vocals, Sound Effects) into dedicated floating visual containers.
  - Uninterrupted Arrangement: Mix and tweak levels or pan inside these floating strips without altering, cluttering, or switching your main arrangement and project screen sets.




**CAUTION:** Despite the floating mixer being highly optimized, opening an excessive number of windows can have a significant impact on CPU performance.

## 3. EDIT MODES & TIMELINE BEHAVIORS

**CAUTION:** Differently than in Pro Tools, Slip/Grid and Shuffle/Spot visual indicators can be lit together.

### Grid and Nudge Value Selection

- **Grid Display Control:** Toggles the visibility of the grid lines on and off in the Edit Window without changing the current Edit Mode. When enabled (highlighted in green), vertical grid

lines are visible across all tracks; when disabled, the grid lines are hidden, allowing for a cleaner visual layout while maintaining grid-snapping if Grid Mode is active.

- **Shortcut:** Cmd + Opt + G (macOS) / Ctrl + Alt + G (Windows).
- **Nudge Indicator and Selector Set:**
  Located directly on the main transport bar for constant visual feedback. It features on-click selector menus and mousewheel selection to quickly cycle through grid/nudge values and time formats (including Milliseconds, Beats, and Samples).
  Nudge Operations: Moving items or selections via Nudge is mapped directly to the + and - keys on the numeric keypad, allowing for fast, hardware-based alignment workflows.




## Grid and Relative Grid Modes

- **Grid Mode Toggle:** Clicking the Grid button locks all editing operations, item movements, and selections to the active grid division. Clicking the button sequentially switches the timeline into Relative Grid mode, allowing items to retain their original offset relative to the nearest grid line when moved. The button color updates dynamically to indicate which grid sub-mode is active.
- **Shortcut:** F4. Press repeatedly to cycle between Absolute Grid and Relative Grid.

## Shuffle Mode

- **Dual-Stage Shuffle Operations:** Clicking the Shuffle button implements a magnetic timeline behavior to close up gaps automatically when items are cut or deleted.
  - **First Click (Per-Track Shuffle):** Activates local ripple editing. Moving or deleting an item affects only the timeline layout of that specific track.
  - **Second Click (All-Tracks Shuffle):** Activates global ripple editing. Deleting a time selection or item shifts the content of all tracks across the entire session to close the gap. Markers will also follow the edit depending on the state of the "Markers follow edit" button on the transport.
- **Shortcut:** F1

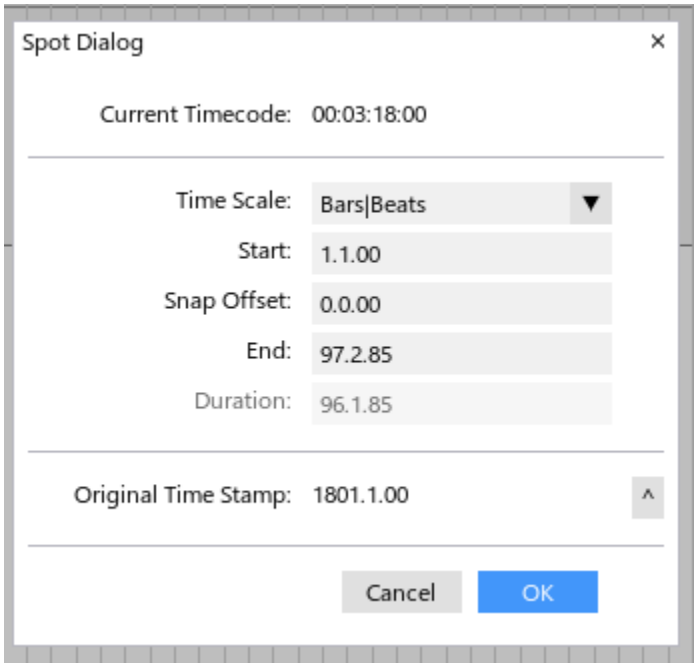
## Slip Mode

- **Slip Mode:** Activates free-form editing, allowing items to be trimmed, moved, or cut smoothly across the timeline without snapping to any temporal grid or adjacent items.
- **Shortcut:** F2

## Spot Mode

- **Spot Mode:** Activates precise placement dialogs. Clicking on any audio item, dragging a file from the Clip List, importing audio via menus, or dragging a file directly from the computer desktop triggers the custom "Spot Dialog" popup window. You can type in the exact target Timecode, Measure, Sample, or Snap Offset to position the asset instantly. It supports manual numeric entries for target parameters or snapping items to their absolute bounds.
- **Shortcut:** F3




## Pre/Post Roll Management

- **Pre/Post-Roll Control:**
  The Pre-Roll and Post-Roll functions are accessible directly on the main transport bar. Users can toggle these states on or off or modify their values with a simple click, enabling quick adjustments to pre- and post-playback behaviors during recording or auditioning without opening secondary configuration menus.The format of the values changes dependently on what is set in the ruler.



# 4. EDITING TOOLS & WORKFLOW PROCESSING

## Smart Tool Architecture

- **Contextual Smart Tool:** Combines the Trim, Selector, and Grabber tools into a single operational interface. The tool changes its function automatically based on where you position the mouse cursor on the vertical axis of the track lane.
- **Single Tools Mode:** Dedicated transport buttons allow you to disable the Smart Tool and lock the mouse cursor into a single dedicated mode when performing repetitive tasks: Trim, Selector, or Grabber.
- **Pencil Tool:** Used to draw freehand automation points. No other functions are assigned to the mouse pointer while active. It locks mouse movement to draw automation curves smoothly without manual node creation.
- **Scrubber Tool:** Allows you to "scrub" across audio tracks by dragging the mouse, playing back the audio at varying speeds to locate specific sounds or edit points. It works per Item, per track, allowing to scrub single items on different tracks.
- **Global Tool Shortcuts:** Defined configuration for lightning fast recall across all tool types:
    - **Zoomer Tool:** F5
    - **Trim Tool:** F6
    - **Selector Tool:** F7
    - **Grabber Tool:** F8
    - **Scrubber Tool:** F9
    - **Pencil Tool:** F10
    - **Smart Tool:** Alt + F8



## Precision Timeline Editing

- **Tab to Transient:** Toggles the behavior of the Tab key via a dedicated visual button on the transport housing.
- **Transient Selection Extensions:** Holding Shift while using the Tab to Transient command extends the selection boundary from the current edit cursor position to the subsequent audio transient ahead or behind.
- **Insertion Follows Playback:** Mapped to the N key to toggle cursor stop preferences. It switches between a tape-style pause (cursor rests exactly where stopped) and returning automatically to the start of your edit selection.

## Link and Synchronization Options

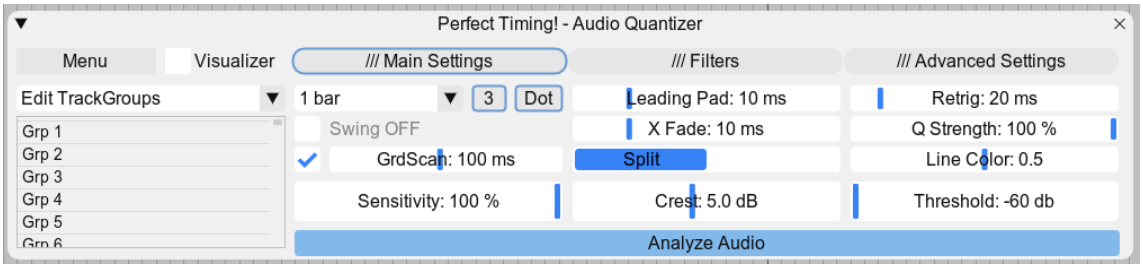
- **Link Track and Edit Selection:** Links track selection to item focus. When disabled, clicking or editing an item on the timeline leaves the track highlight state unchanged in the mixer, preventing accidental parameter alterations. (Shortcut: Shift + T)
- **Link Timeline and Edit Selection:** Connects or decouples time selection from the playback cursor. When enabled, selecting a clip automatically defines the playback range. When disabled, you can perform precision edits or loops on separate sections of the timeline while playback continuously loops on an entirely independent ruler section. (Shortcut: Shift + /)
- **Mirrored MIDI Editing:** Mirrors MIDI data across identical clips, so editing one MIDI clip automatically updates all other copies of that same clip in the session.
  When On, all created midi item acts
  in this way.
- **Automation Follows Edit:** Ensures that automation data breaks, moves, or copies along with the audio or MIDI items when they are edited or relocated.
- **MIDI Merge / Overdub:** Allows you to overdub MIDI data, blending newly recorded MIDI notes with existing notes on the same track instead of overwriting them. In Pro Per, MIDI/Instrument tracks are created with Replace mode enabled by default; this transport button switches the mode to Overdub for all MIDI tracks.

## Fade and Automation Operations

- **Keyboard Focus Fades:** Handled instantly via standard single-key commands without opening secondary dialog boxes. Key D creates a fade-in from the item start to the edit cursor. Key G creates a fade-out from the edit cursor to the item end. Key F draws a seamless crossfade across the intersection of two adjacent, overlapping items.
- **Automation Trim:** When hovering in the upper half of an automation envelope lane, the mouse pointer activates a Trim tool. Dragging vertically moves the selected envelope segment up or down, creating four boundary breakpoints automatically.
- **Clip Gain Mechanics:** Managed via a dedicated knob on the item interface using standard Pro Tools shortcuts. The clip gain envelope view can be toggled via shortcuts specifically for active selections rather than globally, allowing users to bypass clip gain operations without losing underlying written volume nodes.

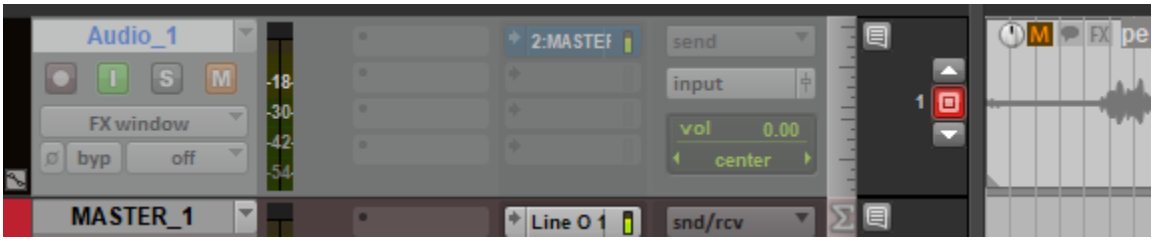
## Beat Detective Integration

- **Perfect Timing (mpl Script):** Integrated and mapped directly to the Pro Tools Beat Detective shortcut profile (Cmd + 8 / Ctrl + 8). It matches the layout and functions of the original, providing transient detection, slicing, grid quantization, and automatic crossfade gap smoothing within a centralized window.

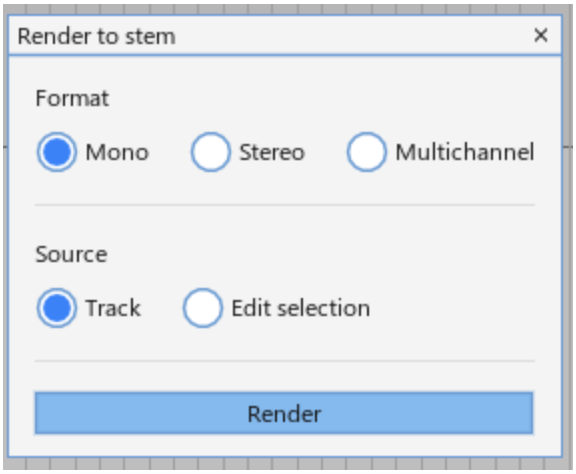



## Performance Optimization and Inactive States

- **Track State Management (Make Inactive):** Tracks can be taken completely offline using a "Make Inactive" command. This state unloads all inserted plugins from CPU threads, sets parameters offline, and updates the layout to a grayed-out, cross-hatched visual scheme. It retains full track positioning and lane assignments, allowing instant re-activation via the track manager or right-click options.




- **Track Commit (Render Tracks):** Advanced track consolidation option. Users can execute a localized track render binding processing effects to a new audio file across either the entire timeline length or an isolated Time Selection. Upon generation, the source track automatically converts to the "Make Inactive" state, archiving the original production chain while preserving processing headroom.







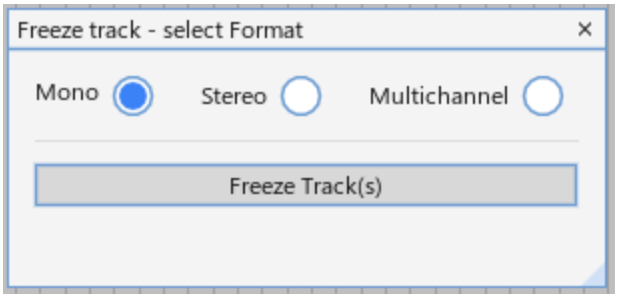
- **Track Freeze:**

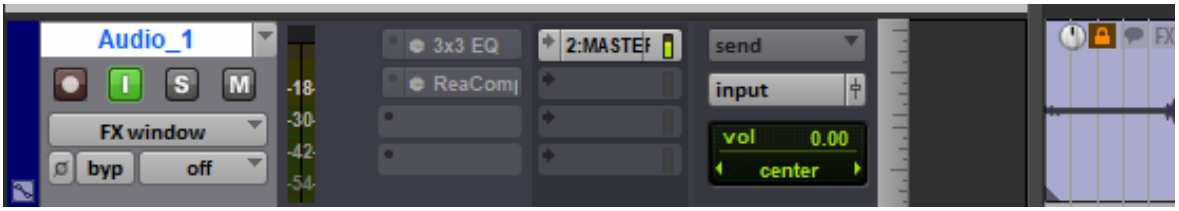
  Enhanced freeze option that leaves in place frozen plugins and shows a clear visualization of the frozen tracks, within the FX slots, even when no plugin was present and there were only edits on the items:when a track contains no track-level inserts (empty FX chain), the custom freeze engine utilizes the empty FX slots as a dynamic status display. The system scans the timeline of the target track and automatically deploys specific text string tags alongside the custom snowflake indicator to represent the underlying non-destructive asset structure:

  - `Ed` **(Edits):** Indicates that the track contains multi-item cuts, splits, or crossfades.
  - `fx` **(Item FX):** Indicates that native Take FX or processing units are applied directly to individual audio items.
  - `env` **(Item Envelopes):** Indicates that active automation's item envelopes are present.

  An additional snow flakes mark on plugins, added for each freeze effectuated, give to you a more clear idea of the order of freezing/unfreezing.

  Self-explanatory custom "Unfreeze last Frozen" and "Unfreeze All" in the menu.

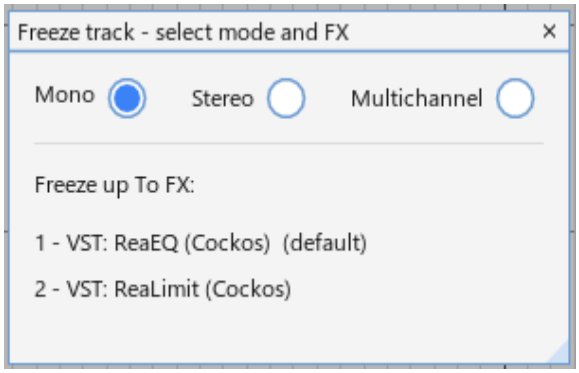


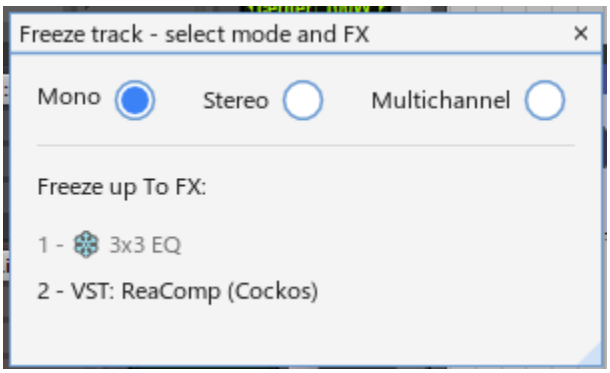




- **Freeze up to insert:**

  Freeze custom option accessible directly from right clicking on fx slots to freeze until a certain plugin ( or all ). A custom, clear window will open, listing all plugins on tracks for fast selection. Same clear visual feedback as for Track Freeze.

  Self-explanatory custom “Unfreeze last Frozen” and “Unfreeze All” in the menu.




**CAUTION:** Never remove manually a frozen plugin or a frozen track indicator. This compromise the correct unfreezing flow.

- **Glue items with handles**

In addition to the standard "Glue items" (consolidate) command via Shift+Alt+3, two new features have been introduced: "Glue items with handles" (2 seconds) mapped to Shift+Alt+2, and "Glue multiple items with handles" mapped to Shift+Alt+1. The latter is particularly useful for item effects processing, serving as an alternative to Pro Tools' AudioSuite effects.

# 5. TEMPO & MONITORING WORKFLOWS

## Tempo Management & Conductor Logic

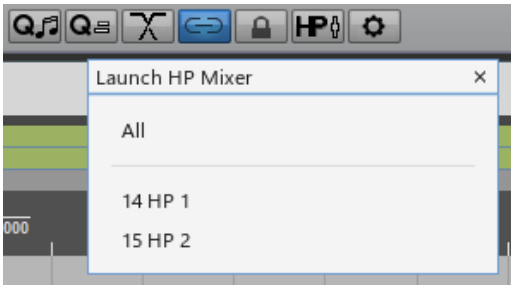
- **Conductor Button Operations:** Toggles the session timebase between following the global Tempo Map or running on an isolated Manual Tempo setting.
- **Conductor Icon Notification:** When the session contains pre-written time signature or the Conductor button is activated, a green Conductor Icon illuminates above the Conductor button and adjacent to the BPM read-out window. This acts as a safety lock, preventing users from altering the global tempo track accidentally from the main box, Tap and Tempo envelope's vertical drag, when Conductor is Off. Clicking this indicator opens an input allowing the engineer to input a manual testing BPM for temporary listening without altering the underlying tempo change points or envelope.

  When the Conductor is On, the green Conductor Icon input box, applies the change to initial tempo point, maintaining relatively all your tempo change points as well the time signatures.

  Re-activating the conductor button from off state re-syncs the timeline to the original map if one was existing.




- **HP Mixer:** A dedicated "HP" button on the main transport housing opens a window from which you can open an isolated headphone send mixer panel. This after you choose from a clear menu the Master HP receiving from the sends you want to operate. A panel maps active artist cue paths into full-throw long faders, allowing quick monitoring adjustments without cluttering screen sets or breaking main mixing views.






## 6. KNOWN BETA LIMITATIONS (V1.7.83 BETA)

- **Track with Control Track Movement Undo Key Command Restriction:** Due to the complex magnetic synchronization of the hidden Control Tracks, the standard keyboard shortcut **Cmd + Z (macOS) / Ctrl + Z (Windows)** is temporarily interrupted exclusively for the vertical dragging action of a track.***Workflow Resumption:*** For all subsequent actions executed *after* the track movement (such as editing, automation, mixing, etc.), the key command Undo system returns to normal operation. ***Recovery Procedure:*** To revert a track displacement, the user can either manually drag the track back to its original position or, in dense sessions, open REAPER's native **Undo History** window and click directly on the previous state with the mouse for a safe real-time realignment of the routing structure.

- **Hardware Control Surface & Hidden Layer Integration Limitations:** Pro Per features bi-directional track selection synchronization; selecting a track within a custom Floating Mixer automatically updates the track focus inside REAPER's native Arrange and Mixer views. Regarding third-party hardware control surfaces (e.g., iCON, Mackie) operating under

MCU/HUI protocols, the system introduces a structural mapping constraint: controllers can exclusively detect, banking-assign, and manipulate tracks that are actively set to visible within REAPER's native TCP or MCP layouts. Consequently, when Control Tracks are hidden, they remain entirely invisible to hardware physical faders. To resolve this constraint and control a specific send, sidechain, or headphone cue master via physical hardware sliders, the user must first expose the hidden control layer globally using the Ctrl+W or Ctrl+E shortcuts. Once the Control Tracks are displayed on-screen, the hardware surface will instantly detect the architecture and follow the floating selection focus.Please note that this configuration DOES NOT include pre-mapped, plug-and-play integrations for third-party hardware control surfaces (such as iCON, Mackie, or SSL).

- **Multi-Stage Undo for Folder Unpacking:** Reverting a track folder structure that was originally generated using drag-and-drop gestures requires 2 or 3 sequential "Undo" keystrokes to return the child tracks back to their original root hierarchy.